This is our **Security Group design for the CityMart 3-tier architecture**

We created **5 security groups**, each mapped to a tier:

Identified Security Groups

- Frontend ALB SG
- Frontend Server SG
- Backend ALB SG
- Backend Server SG
- Database SG

Traffic Flow

```
Internet → Frontend ALB → Frontend EC2 → Backend ALB → Backend EC2 → Database
```

And our security groups enforce **this exact chain**.

1. Frontend ALB (Public Entry Point)

Inbound

- HTTP (80) → `0.0.0.0/0`
- HTTPS (443) → `0.0.0.0/0`

Meaning:

- Public internet can access our app

Outbound

- All traffic → `0.0.0.0/0`

ALB can forward requests anywhere (needed)

2. Frontend EC2 (Web Tier)

Inbound

- HTTP (80) from **Frontend ALB SG**
- Not open to internet
- Only ALB can talk to it

Outbound

- All traffic → `0.0.0.0/0`

Needed to:

- Call backend
- Fetch updates via NAT

Perfect zero-trust layering

3. Backend ALB (Internal Load Balancer)

Inbound

- HTTP (80) from **Frontend Server SG**

Meaning:

- Only our frontend servers can call backend

Internal-only communication

Outbound

- All traffic → `0.0.0.0/0`

Needed for routing to backend instances

4. Backend EC2 (Application Tier)

Inbound

- HTTP (80) from **Backend ALB SG**

Again:

- No direct access
- Only ALB can reach backend

Outbound

- All traffic → `0.0.0.0/0`

Required for:

- DB calls
- External updates via NAT

5. Database (RDS Layer)

Inbound

- MySQL (3306) from **Backend Server SG**

Strict isolation:

- Only backend can access DB
- Frontend cannot access DB directly

Outbound

- All traffic → `0.0.0.0/0`

Default (harmless for RDS)

Architecture Security Pattern We Achieved

We implemented **TRUE tier isolation**:

Layer	Who can access
ALB (public)	Internet
Frontend EC2	Only ALB
Backend ALB	Only frontend
Backend EC2	Only backend ALB
Database	Only backend

Expert-Level Observations

What we did:

- Security group chaining (best practice)
- No direct public access to EC2
- Proper port scoping
- Layer isolation enforced
- Zero trust architecture pattern

Summary

We have correctly implemented:

- Least privilege access
- Layer isolation
- Internal vs external traffic separation
- ALB-based routing security

Security Groups (6) Info

Actions | Export security groups to CSV | Create security group

Find security groups by attribute or tag

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Security group ID	Security group name	VPC ID	Description
sg-0d0d4459b627ee857	citymart-frontend-alb-sg	vpc-0206bba22a724b50b	Citymart Front
sg-0acaf89ebd2aec45b	citymart-database-sg	vpc-0206bba22a724b50b	Citymart Datal
sg-09cfcbc57c3df78b3	citymart-backend-alb-sg	vpc-0206bba22a724b50b	Citymart Back
sg-06be42004925943ad	default	vpc-0206bba22a724b50b	default VPC se
sg-0669523cabfe6be8a	citymart-frontend-server-sg	vpc-0206bba22a724b50b	Citymart Front
sg-0194fdd9bc542789f	citymart-backend-server-sg	vpc-0206bba22a724b50b	Citymart Back

cba_admin

VPC > Security Groups > sg-0d0d4459b627ee857 - citymart-frontend-alb-sg > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
sgr-00f56642c422083e7	HTTP	TCP	80	Cus...	0.0.0.0/0	from internet	Delete
sgr-0898401b1a0eb463d	HTTPS	TCP	443	Cus...	0.0.0.0/0	from internet	Delete

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VPC > Security Groups > sg-0d0d4459b627ee857 - citymart-frontend-alb-sg > Edit outbound rules

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Destination Info		Description - optional Info	
sgr-0f952d808b6120151	All traffic	All	All	Cus...	0.0.0.0/0		Delete

cba_admin

VPC > Security Groups > sg-0669523cabfe6be8a - citymart-frontend-server-sg > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
sgr-0348f829ef0c0de3d	HTTP	TCP	80	Cus...	sg-0d0d4459b627ee857	from frontend load balar	Delete

cba_admin

VPC > Security Groups > sg-0669523cabfe6be8a - citymart-frontend-server-sg > Edit outbound rules

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Destination Info		Description - optional Info	
sgr-088b0c259cf605b43	All traffic	All	All	Cus...	0.0.0.0/0		Delete

cba_admin

VPC > Security Groups > sg-09cfcbc57c3df78b3 - citymart-backend-alb-sg > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
sgr-029cb5e6adec3aaef	HTTP	TCP	80	Cus...	sg-0669523cabfe6be8a	from frontend servers	Delete

Add rule

Cancel Preview changes Save rules

VPC > Security Groups > sg-09cfcbc57c3df78b3 - citymart-backend-alb-sg > Edit outbound rules

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Destination Info		Description - optional Info	
sgr-0b385c2541272972e	All traffic	All	All	Cus...	0.0.0.0/0		Delete

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VPC > Security Groups > sg-0194fdd9bc542789f - citymart-backend-server-sg > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
sgr-0fcc72341b785a0f3	HTTP	TCP	80	Cus...	sg-09cfcbc57c3df78b3	from backend alb	Delete

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VPC > Security Groups > sg-0194fdd9bc542789f - citymart-backend-server-sg > Edit outbound rules

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Destination Info		Description - optional Info	
sgr-01f3d035e6d06043e	All traffic	All	All	Cus...	0.0.0.0/0		Delete

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VPC > Security Groups > sg-0acaf89ebd2aec45b - citymart-database-sg > Edit inbound rules

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info		Description - optional Info	
sgr-06a85a63de66891e1	MYSQL/Aurora	TCP	3306	Cus...	sg-0194fdd9bc542789f	from backend server	Delete

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Destination Info		Description - optional Info	
sgr-0bc466bc089bfc8e1	All traffic	All	All	Cus...	0.0.0.0/0		Delete